Exercise 2 on Combinatorial Proof

The following formula reduces to the one of the previous slide when $q = 1$: $\sum_{k=q}^{n} \binom{k}{q}\binom{n}{k} = \binom{n}{q} 2^{n-q}$, $1 \leq q \leq k \leq n$

Show by combinatorial proof that both sides count the same number in different ways. Hint: You can use the same reasoning as previous slide, but now there are q chairpersons.

LHS: counts the number of ways of selecting a subset with k ($q \leq k \leq n$) out of n elements, and marking q elements among the k selected. Using the terminology of the previous slide, the committee has q chairpersons and any other number of members between zero and $n - q$.

RHS: counts the number of ways of marking q elements out of n (i.e., chairpersons) and choosing k among the remaining $n - q$ elements (to form the rest of the committee).